

The plants sitting where it gets hotter

The estimand

The share of total thermal generating capacity, in MW, across the countries in module 01's hazard table, sitting in a country whose projected count of days at or above a 35C heat index rises by more than k days from baseline (1995-2014) to mid-century (2040-2059) under SSP2-4.5.

Why capacity, not plant count

A grid with one 2 GW coal plant and a grid with twenty 100 MW peakers are not the same exposure. Weighting by MW is the only version of this question a risk manager can use, and the bootstrap below shows how much that weighting costs in precision once a few large plants dominate the total.

The answer

At k=2 days: 43.2% of 260,626 MW
of thermal capacity across 11 countries sits in a country crossing
that threshold -- interval [36.9%, 49.5%] from resampling
plants, 17.9% of the exposed total held by just the
ten largest exposed plants.

Method, and the resolution this was narrowed to

Scope, narrowed on 21/09/2026

The original ask was plants joined to hazard grid cells. The World Bank CCKP API returned HTTP 502 when that build ran, and module 01's committed output is country-level, not gridded, so the gridded version cannot be built from anything currently in hand. This module joins at country resolution and says so on every page rather than waiting on an outage. The gridded version is module 11.

Data

Plants: WRI Global Power Plant Database, output_database csv, fetched live from raw.githubusercontent.com by src/hazard_exposure.py, no key, no transcription.

Hazard: modules/01-heat-stress-gradient/results/heat_gradient.json, already in this repository -- the reuse the roadmap intends, module 03 consumes module 01's output rather than re-fetching the hazard layer.

Join

Filter WRI to thermal fuels (coal, oil, petcoke, plus gas by default -- see the sensitivity page), sum capacity by ISO3, join to the hazard table by ISO3. Every plant in a country inherits that country's national hazard figure; there is no finer join available at this resolution.

Result: the threshold sets the answer

Which countries cross $k=2$

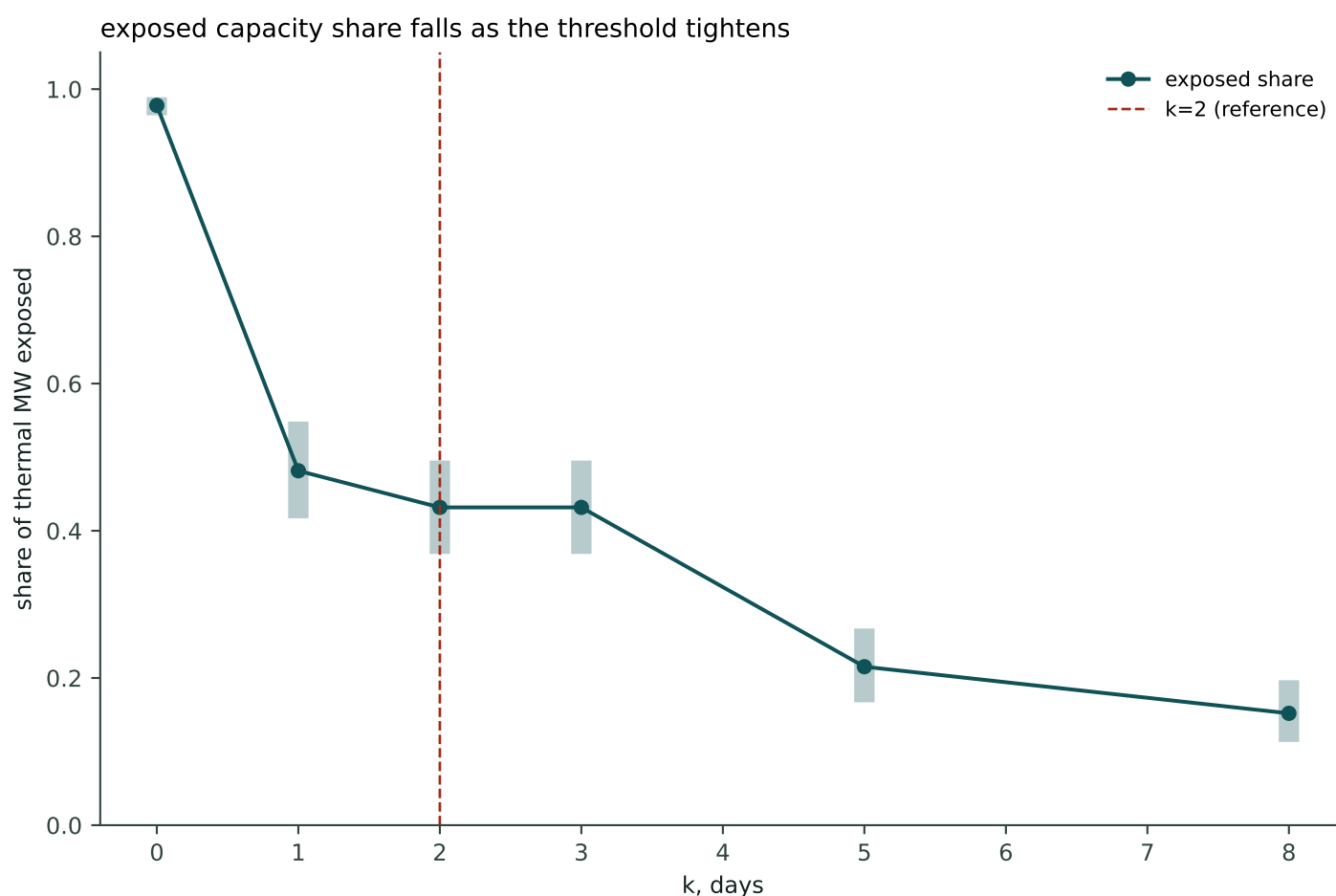
ITA (+3.2d), ESP (+11.6d), GRC (+7.3d), PRT (+6.8d).

4 of 11 countries, holding 112,528 of 260,626 MW.

The interval widens with concentration

At $k=2$, 17.9% of the exposed MW sits in

just ten plants. That is why the bootstrap interval is wide relative to the point estimate: resampling plants, not countries, is what makes a handful of large assets move the answer, and the interval reports that honestly instead of hiding it in a single number.



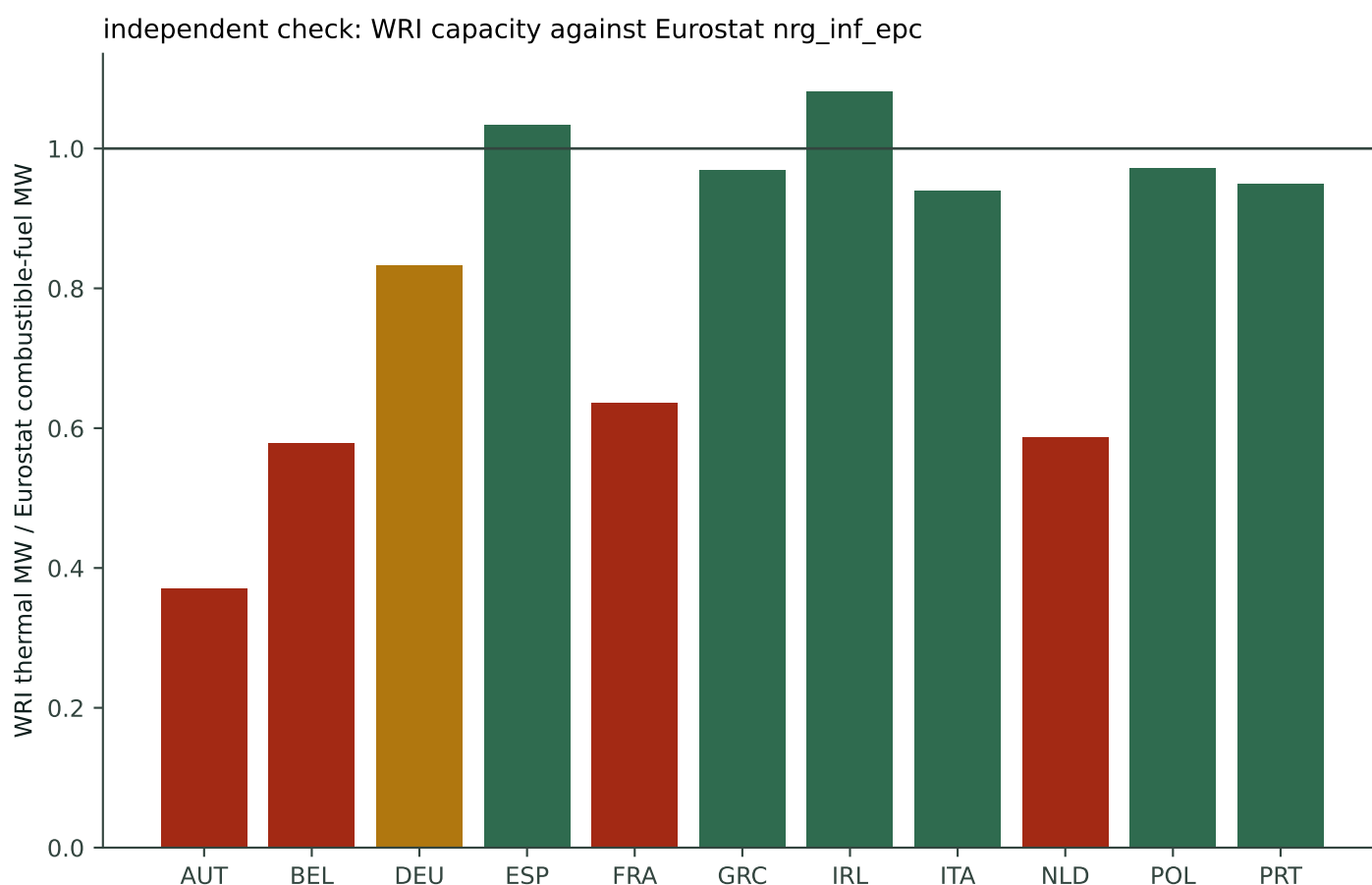
Validation against a source this module did not fit

The check

Eurostat nrg_inf_epc publishes net electrical capacity by fuel group per member state, independent of WRI's asset-level compilation. Combustible-fuel capacity (coal, oil, gas, biofuels and waste combined) is the closest published analogue to this module's thermal definition, so the two are compared directly rather than asserting the asset-level total is right.

What it shows

Spain, Greece, Poland, Portugal and Italy -- the countries that decide the headline number -- sit within 10% of the Eurostat figure. Austria (37%) and Belgium (58%) are the worst outliers: WRI is missing or under-recording thermal capacity there, most likely smaller or newer plants outside GEODB's and Wiki-Solar's typical coverage. Neither country crosses $k=2$, so this does not move the headline, but it is exactly the kind of gap that would matter for a country nearer the threshold.



Sensitivity: the fuel definition, not the threshold

The choice nobody can justify from first principles

Is gas 'thermal'? Engineering says yes -- a CCGT burns fuel to make steam and spin a turbine. Transition-plan language often treats gas as a bridge fuel, distinct from coal. Both readings are defensible, so both are reported rather than one being asserted as correct.

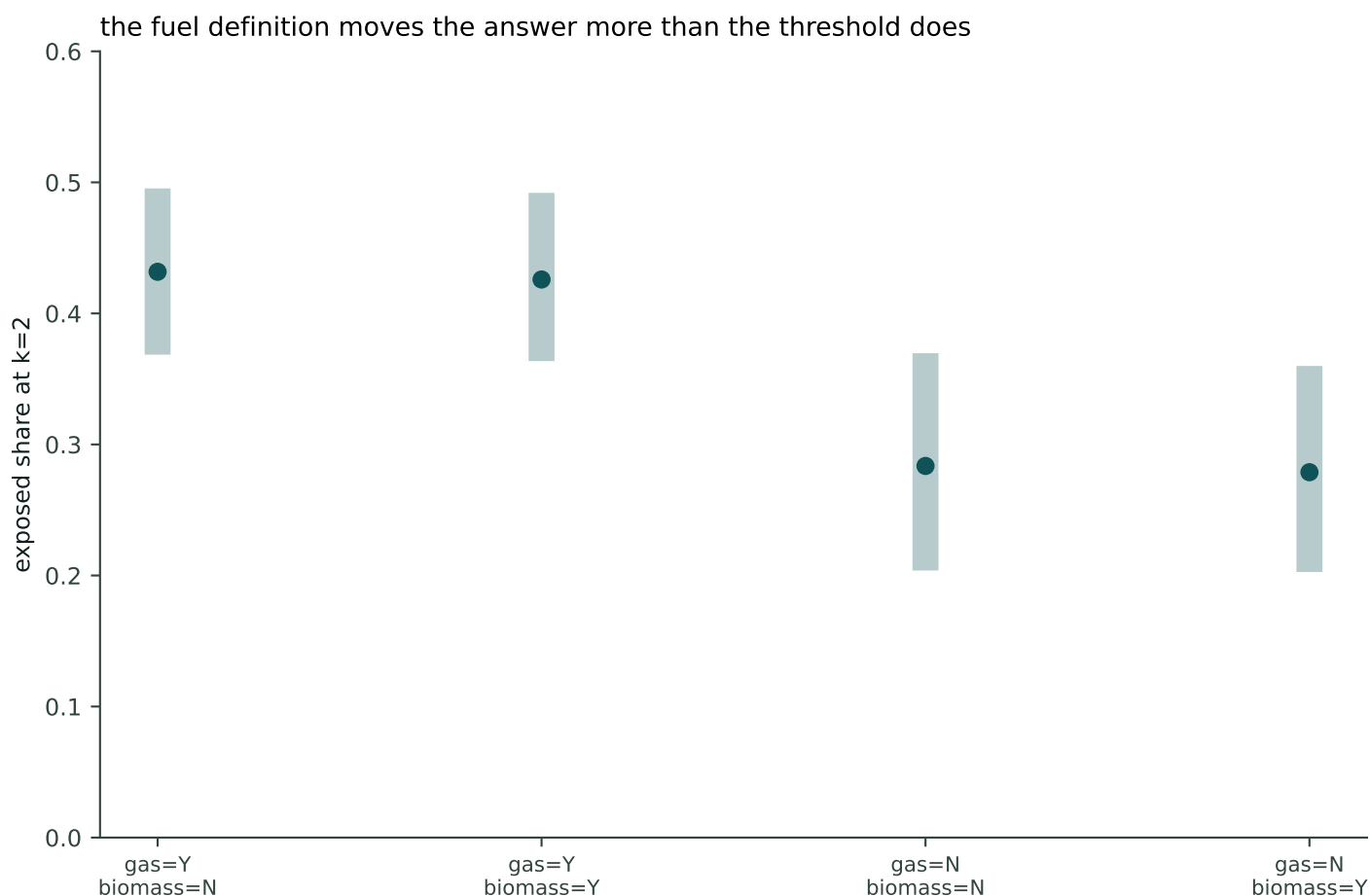
What moves

With gas: 43.2% exposed [36.9%, 49.5%], 260,626 MW in scope.

Without gas: 28.4% exposed [20.4%, 37.0%], 136,693 MW in scope.

Dropping gas removes 123,932 MW of capacity and moves the exposed share by 14.8% points -- more than moving k from 1 to 3 does (48.2% to 43.2%, 5.0% points).

Biomass and waste barely move it either way: both are a small share of installed MW in this country set.



Limits, and what I would build next

The assumption that breaks the conclusion

Country resolution attributes a single national hazard figure to every plant in that country. For Spain this is close to defensible -- one country, one climate zone dominating its thermal fleet. For Germany or France it is close to meaningless: a plant on the Mediterranean coast and one on the North Sea get the same number. This module is least defensible for large, climatically diverse countries, and DEU and FRA -- 28.8% and 5.0% of total thermal MW -- are exactly the countries where that matters most.

No backtest, and why not one for this module

STANDARD.md requires backtesting anything forward-looking. The forward-looking quantity here is CCKP's CMIP6 projection of `change_days`, produced by module 01 and consumed unchanged -- there is no held-out year to test it against, because mid-century has not happened and the projection was never fitted by this module. Module 01's own limitation carries forward instead: ensemble median only, no spread, so this module has no view on the tail of the hazard estimate itself.

The join has no distance decay

A plant at the border of an exposed country and one at the geographic centre get identical treatment. The Eurostat validation on the previous page shows the asset register itself is incomplete for at least two countries, which compounds the resolution problem rather than being independent of it.

Next

Module 11 is this module rebuilt at grid-cell resolution once the CCKP API answers again, with plant coordinates matched to cells under a stated radius and a sensitivity sweep over that radius. The comparison against this module's country-level numbers is itself the finding: it measures what the coarse resolution cost.